

Set Up a Mail Server on Linux

A mail server is a computer on the network that acts as a virtual post office for emails. This article explains how to set up an email server on Gentoo Linux using Postfix and Dovecot. You can choose any other Linux distro for this purpose, provided you make the appropriate modifications, of course.

Postfix is a Mail Transfer Agent (MTA). It waits for mail from Mail User Agents (MUA) from the local network or the outside world. SMTP is a complex protocol that is used both for sending and receiving mails.

Dovecot is an IMAP/POP server that can be used to store and retrieve mails. IMAP stands for Internet Message Access Protocol and is offered by many existing email services like Gmail, Windows Live Mail, etc. POP (POP3) is the Post Office Protocol that was used before IMAP came into existence.

The primary difference between IMAP and POP is that, in the former, you need not download all messages to the client and you can manage multiple folders on the server. In case of POP, you have just one folder—the inbox—and the client must download all messages. This doesn't mean you can't download messages using IMAP, though. IMAP is just more convenient than POP when it comes to handling mails.

The flow of email

Let me explain the flow of email shown in Figure 1:

- The Internet, obviously, is the public facing side of your server.
- Postfix operates on two protocols, namely Submission and SMTP (Simple Mail Transfer Protocol).
- Submission is the same as SMTP, the only difference being that SMTP operates by default on TCP port 25 while Submission operates on TCP port 587. This enables naive users to send emails via the Submission ports using MUAs because many ISPs block port 25.
- There's also SMTPS—the SSL variant of the SMTP protocol that usually operates on TCP port 465. SMTP is a clear text protocol, and in this case, just the data is encrypted before transmission and decrypted after transmission at the server.
- There are a couple of reasons why the Submission port exists. You can read

more about it at Wikipedia - http://en.wikipedia.org/wiki/Mail_submission_agent

- LMTP (Local Mail Transfer Protocol): This is used by Postfix to transfer mails to the particular account on Dovecot.
- Amavisd, ClamAV and SpamAssassin: ClamAV and SpamAssassin are virus and spam filters, respectively, but in order to use both simultaneously, we need Amavisd, which acts as a forwarder.
- PostgreSQL: This is the database used for storing various

